

Computational Modelling of Grain Growth

The Fan–Chen Continuum Field Model: Physics, Mathematics, and Finite Element Implementation

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1 Physical Picture

A polycrystal is a collection of grains – regions of identical crystallographic orientation – separated by grain boundaries, which carry a positive excess free energy σ (J/m²). This excess energy drives boundary migration to reduce total boundary area (grain growth / coarsening).

Sharp-interface (classical) results

- Local velocity: $v = M\sigma K$, where K is the sum of principal curvatures and M the boundary mobility.

- Mean grain radius: $\bar{R}^2 - \bar{R}_0^2 = kt$ (parabolic growth, exponent $m = 2$).
- 2D topology (von Neumann–Mullins law, with 120° triple junctions):

$$\frac{dA}{dt} = \frac{\pi}{3} M \sigma (n - 6)$$

where n is the number of sides of a grain – grains with $n < 6$ shrink, $n > 6$ grow.

Traditional models (Potts, vertex, boundary-tracking) treat the boundary as a zero-thickness surface and must explicitly track its position and handle topological events (grain deletion, neighbour switching) by hand.

The diffuse-interface alternative

The Fan–Chen model instead gives the boundary a small but finite thickness. Its location is never tracked explicitly – it is simply wherever certain continuous fields vary in space. Topological events (grain shrinkage/disappearance, coalescence) happen automatically as a consequence of the PDE, with no special-case logic required.

2 Order Parameters

The polycrystal is described by p continuous, **non-conserved** fields

$$\eta_1(\mathbf{r}, t), \eta_2(\mathbf{r}, t), \dots, \eta_p(\mathbf{r}, t)$$

called *orientation field variables*. Inside a grain of orientation i : $\eta_i = 1$ (or -1), all other fields $= 0$. Across a boundary, exactly the two competing fields vary smoothly; all others remain zero.

- η_i is a **label**, not a misorientation angle – every boundary in the model is energetically identical.
- **Non-conserved**: there is no constraint $\sum_i \eta_i = \text{const}$. This is what makes the dynamics Allen–Cahn (2nd order PDE) rather than Cahn–Hilliard (4th order) – a major practical simplification for FEM, since only C^0 elements are required.
- Formally $\eta_i \in [-1, 1]$: the free energy (below) has $2p$ degenerate minima at $(\pm 1, 0, \dots, 0)$, etc. A common, harmless simplification is to restrict to $[0, 1]$ (p orientations) by seeding only positive noise.

3 Strong Form

3.1 Free energy functional

$$F[\boldsymbol{\eta}] = \int_{\Omega} \left[f_0(\boldsymbol{\eta}) + \sum_{i=1}^p \frac{\kappa_i}{2} |\nabla \eta_i|^2 \right] dV$$

$$f_0(\boldsymbol{\eta}) = \sum_{i=1}^p \left(-\frac{\alpha}{2} \eta_i^2 + \frac{\beta}{4} \eta_i^4 \right) + \gamma \sum_{i=1}^p \sum_{j>i}^p \eta_i^2 \eta_j^2$$

with $\alpha, \beta, \kappa_i, L_i > 0$.

Condition on γ : $\gamma > \beta/2$.

Writing $S = \sum_i \eta_i^2$, $Q = \sum_i \eta_i^4$, one finds $f_0 = -\frac{\alpha}{2}S + (\frac{\beta}{4} - \frac{\gamma}{2})Q + \frac{\gamma}{2}S^2$. If $\gamma < \beta/2$ the system prefers to *spread* order across many fields (unphysical); if $\gamma = \beta/2$ the minima merge into a degenerate ring/sphere (no distinct grains); only $\gamma > \beta/2$ gives $2p$ isolated, degenerate wells at $(\pm 1, 0, \dots, 0), \dots$ – one per orientation.

3.2 Functional derivative

$$\frac{\delta F}{\delta \eta_i} = \frac{\partial f_0}{\partial \eta_i} - \kappa_i \nabla^2 \eta_i, \quad \frac{\partial f_0}{\partial \eta_i} = -\alpha \eta_i + \beta \eta_i^3 + 2\gamma \eta_i \sum_{j \neq i} \eta_j^2$$

3.3 Governing PDE (Allen–Cahn / TDGL)

Assuming a diagonal mobility matrix $L_{ij} = L_i \delta_{ij}$:

$$\frac{\partial \eta_i}{\partial t} = -L_i \left[-\alpha \eta_i + \beta \eta_i^3 + 2\gamma \eta_i \sum_{j \neq i} \eta_j^2 - \kappa_i \nabla^2 \eta_i \right] \quad i = 1, \dots, p$$

A system of p coupled, semilinear, second-order parabolic PDEs. **Coupling between fields is purely algebraic** (through $\sum_{j \neq i} \eta_j^2$) – there is no gradient cross-coupling. This is structurally important for both stability analysis and FEM assembly (Sec. 5).

Boundary conditions: periodic, or homogeneous Neumann $\nabla \eta_i \cdot \mathbf{n} = 0$ (zero flux). Both are the *natural* BC of the weak form (Sec. 4) – no flux enters or leaves.

Initial condition: small random noise, e.g. $\eta_i^0(\mathbf{x}) \in [-0.001, 0.001]$ i.i.d. at every node – mimics an undercooled liquid with no crystalline order.

4 From Strong Form to Weak Form

Multiply by test function $w_i \in H^1(\Omega)$ and integrate over Ω :

$$\int_{\Omega} \frac{\partial \eta_i}{\partial t} w_i dV = -L_i \int_{\Omega} \frac{\partial f_0}{\partial \eta_i} w_i dV + L_i \kappa_i \int_{\Omega} (\nabla^2 \eta_i) w_i dV$$

Integrate the Laplacian term by parts (Green’s identity):

$$\int_{\Omega} (\nabla^2 \eta_i) w_i dV = \underbrace{\int_{\partial \Omega} w_i (\nabla \eta_i \cdot \mathbf{n}) dS}_{=0 \text{ (natural BC)}} - \int_{\Omega} \nabla \eta_i \cdot \nabla w_i dV$$

Why the boundary term vanishes

- **Neumann:** $\nabla \eta_i \cdot \mathbf{n} = 0$ pointwise everywhere on $\partial \Omega$ – each term is individually zero.
- **Periodic:** the two opposite-face contributions are equal in magnitude, opposite in sign (outward normals point opposite ways; η_i and its normal derivative match by the periodicity condition), and cancel *term by term*. Equivalently: on the torus $\partial \Omega = \emptyset$ – there is no boundary to integrate over. This is the same interior-facet cancellation that occurs between every pair of adjacent elements in any FEM mesh; periodicity simply identifies the last element’s outward face with the first element’s, turning an exterior facet into an interior one.

4.1 Weak form

Find $\eta_i(\cdot, t) \in V = H^1(\Omega)$ (or H_{per}^1) such that $\forall w_i \in V$:

$$\int_{\Omega} \frac{\partial \eta_i}{\partial t} w_i dV + L_i \kappa_i \int_{\Omega} \nabla \eta_i \cdot \nabla w_i dV + L_i \int_{\Omega} \left[-\alpha \eta_i + \beta \eta_i^3 + 2\gamma \eta_i \sum_{j \neq i} \eta_j^2 \right] w_i dV = 0$$

Because only $\nabla \eta_i$ (not $\nabla^2 \eta_i$) appears, H^1 suffices – C^0 Lagrange elements are sufficient, no C^1 /mixed formulation is needed (unlike Cahn–Hilliard).

5 Matrix–Vector (Galerkin) Form

Interpolate $\eta_i(\mathbf{x}, t) \approx \sum_a \eta_i^a(t) N_a(\mathbf{x})$, take $w_i = N_a$. Define

$$M_{ab} = \int_{\Omega} N_a N_b dV \text{ (mass matrix),} \quad K_{ab} = \int_{\Omega} \nabla N_a \cdot \nabla N_b dV \text{ (stiffness matrix)}$$

The nonlinear reaction integral has no exact closed form; it is handled one of two ways – **never both**:

Path 1 – direct quadrature	Path 2 – group FE (product approx.)
$F_a^{(i)} = \sum_q N_a(\mathbf{x}_q) \left[\frac{\partial f_0}{\partial \eta_i} \right]_{\mathbf{x}_q} w_q J_q $	$F_i^b = \frac{\partial f_0}{\partial \eta_i}(\eta^b)$ evaluated once per <i>node</i> , no quadrature
Already fully integrated.	Raw nodal values, unintegrated.
Use directly: $\text{RHS} += \mathbf{F}$.	Multiply by M afterward: $\text{RHS} += M * \mathbf{F}$.

Common bug: mixing the two – assembling F via quadrature (Path 1, already integrated) and *then* multiplying by M again. This double-integrates the term and silently corrupts the scheme.

The full semi-discrete (method-of-lines) system, one block per field:

$$M \dot{\boldsymbol{\eta}}_i + L_i \kappa_i K \boldsymbol{\eta}_i + L_i \mathbf{F}_i(\boldsymbol{\eta}) = \mathbf{0}$$

K never mixes fields; only $\mathbf{F}_i$ does, algebraically, node by node.

6 Explicit Time Stepping

Forward Euler applied to the full weak form (matrix form, mass-lumped or via a linear solve):

$$M \eta_i^{n+1} = M \eta_i^n - \Delta t L_i \kappa_i K \eta_i^n - \Delta t L_i M \mathbf{F}_i(\boldsymbol{\eta}^n)$$

i.e. solve $M \eta_i^{n+1} = \text{RHS}$ at every step (an actual linear solve – M is *not* the identity).

Jacobi (synchronous), never Gauss–Seidel (sequential), across fields

At each timestep, **freeze the entire old state η^n** and evaluate $\mathbf{F}_i(\eta^n)$ for *every* i using it. Never use an already-updated η_j^{n+1} while still solving for η_k^{n+1} within the same step (“Gauss–Seidel-in-time”). This breaks the permutation symmetry the continuous PDE has among the p fields (an arbitrary loop order would change the answer), and is unnecessary: since the diffusion operator never couples fields, all p linear solves are algebraically independent once η^n is frozen – they can be solved in parallel, or batched into one multi-RHS solve reusing a single LU factorisation (since M, K are identical for every field when L_i, κ_i are uniform).

7 Choosing the Mesh Spacing h

The equilibrium grain-boundary profile has a finite width, fixed by the model constants alone (independent of domain size):

$$w_{10-90} \sim 5.4\sqrt{\kappa/(2\alpha)} \quad (\text{computed numerically by solving the 1D two-field equilibrium problem})$$

Reference values ($\alpha = \beta = \gamma = 1, \kappa = 2$): $\sigma_{gb} = 0.548$, $w_{10-90} = 5.36$, equipartition of bulk/gradient energy confirmed to solver tolerance.

Rule: choose h so that

$$h \leq \frac{w_{10-90}}{5 \text{ to } 8}$$

Below ~ 2 – 3 elements across the boundary, the discretisation cannot represent the smooth transition at all – the diffuse interface can **lock onto the mesh (“lattice pinning”)** and stop moving entirely, independent of how long the simulation is run. 5–8 elements is the safe range (numerically verified: $h = 1 \Rightarrow 5.4$ elements, 0.6% error in σ_{gb} ; $h = 0.5 \Rightarrow 10.7$ elements, 0.2% error).

Procedure: (1) compute w_{10-90} by solving the 1D equilibrium boundary-value problem for your α, κ ; (2) divide by 5–8; (3) verify by mesh-refinement study that σ_{gb} has converged at the chosen h – do not trust the formula alone.

A related, distinct length scale: linear-instability cutoff

Near the unstable state $\eta = 0$ (the random-noise IC), Fourier modes with wavenumber k grow as $r(k) = L(\alpha - \kappa k^2)$ – unstable for $k < k_c = \sqrt{\alpha/\kappa}$, giving a minimum surviving feature size

$$\ell_c = \frac{2\pi}{k_c} = 2\pi\sqrt{\kappa/\alpha}$$

Because w_{10-90} and ℓ_c scale the *same way* with $\sqrt{\kappa/\alpha}$, pure white-noise nucleation tends to produce grains only a few interface-widths across, *regardless of the absolute value of κ/α* . If nucleated grain size $\sim w$, there is no flat grain interior anywhere – the reaction term (which sharpens) and the gradient term (which smooths) fight to a standstill everywhere at once, producing a frozen, labyrinthine (maze-like) pattern rather than clean polygonal grains, that does not meaningfully coarsen even over very long runs (confirmed by measuring the pattern’s own autocorrelation length ≈ 6.5 , matching $w = 5.36$ almost exactly). **Fix:** decouple nucleation length from κ, α by seeding a Voronoi tessellation (explicit control of initial grain size) instead of relying on white noise.

8 Choosing the Timestep Δt

Depends entirely on whether the reaction term is treated explicitly.

8.1 Von Neumann stability (fully explicit scheme)

Linearising and Fourier-transforming, the amplification factor per mode is

$$G(\mathbf{k}) = 1 - \Delta t L(\kappa \lambda(\mathbf{k}) + \mu)$$

where $\lambda(\mathbf{k}) \geq 0$ is the discrete Laplacian’s eigenvalue and μ the local reaction-Hessian eigenvalue ($\mu = -\alpha$ near $\eta = 0$, unstable; $\mu = +2\alpha$ near a well, stable).

Stencil	$\lambda_{\max}$	
Standard 5-point FD	$8/h^2$	most restrictive
Fan–Chen 9-point FD	$4/h^2$	
Lumped-mass Q1 FEM	$8/(3h^2)$	mildest

$$\Delta t \leq \frac{2}{L(\kappa \lambda_{\max} + A)}, \quad A = \text{bound on } \max(\text{reaction Hessian eigenvalue})$$

Halving h roughly quarters the admissible Δt – the dominant cost of fine meshes with fully explicit stepping.

8.2 Unconditional stability (semi-implicit / linearly-stabilised splitting)

Move a linear term $A\eta_i$ (with $A \geq \max \mu$ everywhere) to the implicit side:

$$[(1 + \Delta t L_i A)M + \Delta t L_i \kappa_i K] \eta_i^{n+1} = M \left[(1 + \Delta t L_i A) \eta_i^n - \Delta t L_i \mathbf{F}_i(\eta^n) \right]$$

Common bug: adding A on the LHS matrix but forgetting to add it back on the RHS. This introduces uncompensated extra damping – observed empirically as a near-constant *geometric* decay factor per step (e.g. $\eta^{n+1} \approx \eta^n / (1 + \Delta t L A)$), collapsing the field toward zero regardless of the true reaction dynamics.

The resulting amplification factor,

$$G(\mathbf{k}) = \frac{1 + \Delta t L(A - \mu)}{1 + \Delta t L(A + \kappa \lambda(\mathbf{k}))} \xrightarrow{\Delta t \rightarrow \infty} \frac{A - \mu}{A + \kappa \lambda(\mathbf{k})} \quad (\text{finite}),$$

is bounded for *any* $\Delta t > 0$ (vs. the explicit scheme’s $G \rightarrow -\infty$) – genuine unconditional stability, empirically confirmed stable at Δt up to $500\times$ the explicit limit.

Unconditional stability only guarantees boundedness, not accuracy. Choose Δt to resolve: (i) the reaction relaxation time $\tau \sim 1/(L\alpha)$; (ii) boundary migration, $\Delta t \ll h/v$ where $v = L\kappa K$.

8.3 Total simulated time – a separate budget from Δt

Δt controls per-step accuracy/stability; $N_T \Delta t$ (total simulated time) controls whether enough physical time has elapsed to see anything at all. Two timescales must both fit:

- **Nucleation/crystallisation:** growth rate near $\eta = 0$ is $r \approx L\alpha$; growing from noise amplitude η_0 to $O(1)$ takes several e -folding times, $t \sim \ln(1/\eta_0)/(L\alpha)$ at minimum – empirically $t \sim 30\text{--}80$ for $\eta_0 = 0.001$, $L\alpha = 1$ (a sharp, almost sudden transition, not gradual).
- **Coarsening:** $\bar{R}^2 - \bar{R}_0^2 = 2L\kappa t$ – growing from $\bar{R} = 10$ to $\bar{R} = 100$ needs $t \sim 2500$.

9 Stability Analysis – Summary

Scheme	Stability behaviour
Fully explicit	Conditionally stable; $\Delta t \leq 2/[L(\kappa\lambda_{\max} + A)]$, scales as h^2 . Violating it produces exponential blow-up (often visible as growth $\sim 10\times$ faster than the physical rate $L\alpha$, or a spike-then-collapse pattern).
Semi-implicit (correctly stabilised)	Unconditionally stable for any $\Delta t > 0$ – amplification factor asymptotes to a finite value rather than diverging. Δt becomes an accuracy choice, not a stability one.
Semi-implicit (RHS add-back missing)	<i>Appears</i> stable (bounded) but exhibits spurious geometric decay toward zero – a subtle, easily-overlooked implementation bug distinct from a true instability.

10 Practical Debugging Checklist

Verify each layer independently before trusting the coupled nonlinear system:

1. M : symmetric, positive diagonal; total mass = domain volume.
2. K : symmetric; $K \cdot \mathbf{1} = \mathbf{0}$ row-sum identity (partition of unity) – *only over unconstrained DOFs*; deal.II intentionally places a placeholder 1 on constrained (slave) diagonal entries, which is correct, not a bug.
3. **Constraints:** correct master/slave pairing; `constraints.close()` called *before* assembly; `distribute()` called on the initial condition and after every solve.
4. **Linear solve:** round-trip test – $x_{\text{known}} \rightarrow Mx_{\text{known}} = \text{rhs} \rightarrow \text{solve} \rightarrow x_{\text{solved}} \approx x_{\text{known}}$.
5. **Reaction term, isolated (“0-D test”):** strip out K (no neighbours) and M (cancels algebraically) entirely; run the literal reaction-term code as a single-scalar recurrence. Stage A: one field only (cross term must vanish identically – checks for a missing “subtract self” bug in $S - \eta_i^2$). Stage B: two competing fields (checks winner-take-all sign/shape).

6. **Sign of the TDGL equation:** $\partial\eta/\partial t = -L\delta F/\delta\eta$ – a dropped minus sign makes $\eta = 0$ artificially *stable*, and noise decays back to zero instead of growing (produces ”boundaries appear, then freeze” within one reaction *e*-folding time).
7. **Diagnostics:** track $\max|\eta|$ (nucleation-completeness only – saturates at ≈ 1 and stays there through all subsequent coarsening, uninformative afterward) and total free energy $F(t)$ (guaranteed monotonically non-increasing for a correct gradient flow; remains sensitive throughout the whole run).